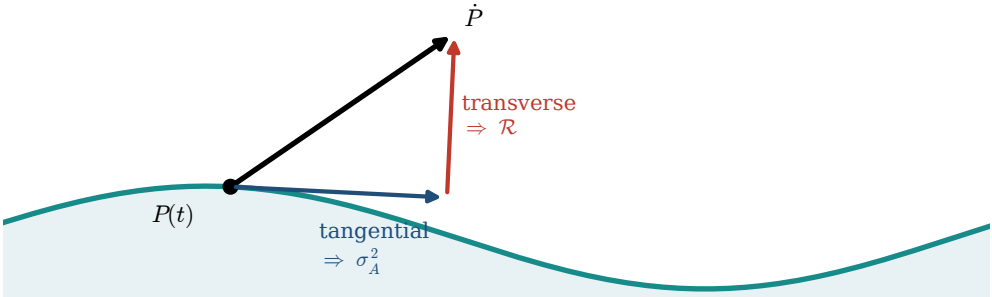
Fisher's assertion: the genotype distribution never leaves $\mathcal{M}$



$\mathcal{M}$: gene frequency product measures

(Hardy Weinberg proportions and linkage equilibrium)